

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Expression of Ideas	Rhetorical Synthesis	Easy

Question

While researching a topic, a student has taken the following notes:

- The Haber-Bosch process is an industrial process used to manufacture ammonia (NH₃).
- It was invented by chemists Fritz Haber and Carl Bosch in 1910.
- The process’s primary reaction combines nitrogen (N₂) from the air with hydrogen (H₂).
- It requires an iron catalyst and high temperatures and pressures.
- Most of the ammonia produced by this process is used in fertilizers.

The student wants to provide an overview of the Haber-Bosch process. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. The Haber-Bosch process needs nitrogen, hydrogen, and an iron catalyst.
- B. The Haber-Bosch process uses an iron catalyst along with high temperatures and pressures to manufacture ammonia from nitrogen and hydrogen.
- C. Chemists Fritz Haber and Carl Bosch invented an industrial process to manufacture ammonia to be used in fertilizers.
- D. In 1910, chemists Fritz Haber and Carl Bosch invented the Haber-Bosch process, which requires high temperatures and pressures.

Correct Answer: B

Rationale

Choice B is the best answer. The sentence provides an overview of the Haber-Bosch process, explaining that it uses an iron catalyst, high temperatures, and high pressures to manufacture ammonia from nitrogen and hydrogen.

Choice A is incorrect. While the sentence identifies some of the process’s required components (nitrogen, hydrogen, and an iron catalyst), the overview is incomplete; it doesn’t explain what the process does (produces ammonia) or mention the conditions needed for it to function (high temperatures and pressures). Choice C is incorrect. While the sentence indicates who invented the process (Fritz Haber and Carl Bosch) and that it produces ammonia, the overview is incomplete; it doesn’t mention any of the necessary ingredients or conditions. Choice D is incorrect. While the sentence mentions the inventors of the process along with the required conditions (high temperatures and pressures), it doesn’t explain that the process produces ammonia or indicate what components are needed.